TAYLOR & FRANCIS

# Marketing ROI Audit

*End-to-End Data Analytics Project Report*

Python · MySQL · Power BI

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# 1. Executive Summary

This report presents the findings of a complete marketing ROI audit for Taylor & Francis, conducted to evaluate the commercial return on a £650,000 marketing investment across four channels and twenty campaigns. A last-touch attribution model was implemented in SQL to assign each published article's APC revenue to the channel that last engaged the author before submission, enabling direct channel-level ROAS comparison.

The audit identified a significant and actionable budget misallocation. The four channels delivered sharply divergent returns:

| Channel | Spend | Attributed Revenue | ROAS | CPA | Conversions |
|---|---|---|---|---|---|
| Email | £50,000 | £1,428,500 | 28.57x | £142 | 352 |
| Webinar Series | £80,000 | £780,500 | 9.76x | £394 | 203 |
| Organic Search | £120,000 | £378,000 | 3.15x | £1,290 | 93 |
| Social Ads | £400,000 | £213,000 | 0.53x | £7,692 | 52 |

Social Ads consumed 62% of the total budget yet returned only 0.53x ROAS - a net loss of £187,000. Email consumed 7.7% of the budget and generated 51% of all attributed revenue at a 28.57x ROAS. The overall portfolio achieved a 4.31x ROAS, exceeding the £2,500,000 revenue target by £300,000 -a surplus driven entirely by Email and Webinar Series, not by the current allocation strategy.

Audience segmentation identified Corporate Pharma Research Scientists as the single most commercially valuable author segment at £677,500 in attributed revenue. K-Means clustering confirmed this segment constitutes approximately 78% of the High Intent Whale cluster - the highest email-engaging behavioural group, directly linking the channel efficiency finding to a specific, targetable audience.

**Primary Recommendation:** Reallocate a minimum of 50% of Social Ads budget toward Email and Webinar Series. Combined with targeted Call for Papers campaigns directed at Corporate Pharma Research Scientists via Email, this reallocation is expected to materially increase overall marketing ROAS. Full supporting analysis and recommendations are in Section 9.

# 2. Project Background and Objectives

## Business Context

Taylor & Francis invested across four marketing channels - Email, Webinar Series, Organic Search, and Social Ads and twenty campaigns to attract academic and corporate authors and drive peer-reviewed article submissions. Each published article generates revenue through an Article Processing Charge (APC), which varies by subject area. The marketing team required an independent audit to determine which investments were generating commercial return and which were destroying value.

## Business Questions

- Which marketing channels are generating positive ROI and which are loss-making?
- How much revenue can be directly attributed to each channel and each campaign?
- Which author segments - by institution type, job title, and behaviour are most commercially valuable?
- Are there distinct behavioural archetypes among authors that should be targeted differently?
- Which subject areas drive the highest submission volume and revenue?
- Which journals have the highest and lowest article acceptance rates?

## Project Scope and Pipeline

The project covers the full analytical pipeline from raw data through to a published interactive report. Three phases connect sequentially - each phase produces outputs consumed by the next.

| Phase | Tool | Key Deliverable |
|---|---|---|
| Phase 1 | Python | ETL, EDA, K-Means Clustering, CSV Export |
| Phase 2 | MySQL | Schema Design, Data Ingestion, Attribution Building |
| Phase 3 | Power BI | Star Schema, DAX Measures, 4-Page Interactive Report |

# 3. Data Overview

The dataset comprises four core relational tables and one derived MySQL view covering authors, marketing interactions, article submissions, and marketing spend. The data spans a full calendar year (January–December 2024) across four channels, twenty campaigns, five countries, four journals, and four subject areas.

## Core Tables

| Table | Records | Key Columns | Description |
|---|---|---|---|
| authors | 5,000 | author_id, job_title, institution_type, country | Author demographics and consent data |
| interactions | 90,090 | interaction_id, author_id, campaign_id, channel | Marketing touchpoint records |
| submissions | 2,550 | submission_id, author_id, journal_name, status, apc_gbp | Article submissions and APC revenue |
| marketing_spend | 20 | campaign_id, channel, campaign_budget_gbp | Campaign budget allocations |
| submissions_attributed (view) | 700 | submission_id, last_touch_channel, apc_gbp | Last-touch attribution output - MySQL view |

*Note: submissions_attributed is a MySQL view built using CTEs and ROW_NUMBER() window functions. It encapsulates the attribution logic and is loaded directly into Power BI as a queryable table.*

## Data Dimensions

| Dimension | Values |
|---|---|
| Channels | Email, Webinar Series, Organic Search, Social Ads |
| Campaigns | 20 campaigns - 5 per channel (CMP_EM_01 to CMP_SO_05) |
| Journals | Journal of Medicine, Journal of Oncology, Journal of Physics, Journal of Social Sciences |
| Subject Areas | Medicine, Oncology, Physics, Social Sciences |
| Countries | United States, United Kingdom, Germany, India, Japan |
| Institution Types | Corporate, Academic, Unknown |
| Author Job Titles | Clinical Researcher, Pharma Research Scientist, AI Researcher, Professor |
| APC Revenue Tiers | £2,500 (Social Sciences), £4,000 (Physics), £4,500 (Medicine), £5,500 (Oncology) |

# 4. Methodology

## 4.1 Last-Touch Attribution Model

Last-touch attribution assigns 100% of a published article's APC revenue to the marketing channel that last interacted with the author before their submission date. The logic was implemented in SQL using two steps:

- Step 1 - Rank Interactions: For each author with a published submission, all marketing interactions that occurred before the submission date were identified and ranked in descending chronological order using ROW_NUMBER() OVER (PARTITION BY author_id ORDER BY interaction_date DESC).
- Step 2 - Select Last Touch: Only rank = 1 interactions were retained and joined to the submissions table to assign channel and campaign attribution to each published article's APC revenue.
- Step 3 - Productionise as View: The complete logic was encapsulated in the submissions_attributed MySQL view, making it immediately available to Power BI as a queryable table.

> **Attribution Assumption:** Last-touch is a conservative model. It credits only the final touchpoint and ignores earlier interactions that may have contributed to the author's submission decision. Channel ROAS values represent a lower-bound estimate for channels that operate earlier in the funnel (e.g. Organic Search).

## 4.2 K-Means Clustering Methodology

K-Means clustering was applied exclusively to corporate authors using two behavioural features: average email opens and average webinar attendance per user. The optimal cluster count (k = 3) was determined using the Elbow Method. Features were standardised using StandardScaler before fitting to prevent scale dominance. The resulting clusters were profiled and validated by merging cluster labels back with author demographic data.

## 4.3 Power BI Data Model

A star schema was built in Power BI Desktop connecting directly to the MySQL database. The model includes five relationships - four active and one inactive (submissions to Date Table, activated via USERELATIONSHIP() in DAX time intelligence measures). 23 DAX measures were created across 6 categories covering revenue, funnel metrics, channel attribution, channel KPIs, time intelligence, and campaign detail.

# 5. Phase 1 - Python: ETL, EDA and Clustering

## 5.1 ETL - Data Cleaning and Transformation

Raw datasets were loaded into Pandas DataFrames for initial health checks. The following issues were identified and resolved:

| Issue Identified | Resolution Applied |
|---|---|
| 250 null values in institution_type column | Retained as 'Unknown' - not imputed, to preserve analytical transparency |
| Inconsistent capitalisation and trailing whitespace in job_title and channel columns | Normalised using .str.strip().str.title() to prevent JOIN failures in MySQL |
| Negative values found in historical_publications column | Identified as illogical data point during health check - flagged for review |
| Mixed date formats in interaction_date - UK (DD-MM-YYYY) and US (MM/DD/YYYY) | Custom parse_interaction_date() function built to handle both formats, standardising to YYYY-MM-DD for SQL ingestion |
| submission_date in DD-MM-YYYY format | Standardised to YYYY-MM-DD using pd.to_datetime with explicit format parameter |

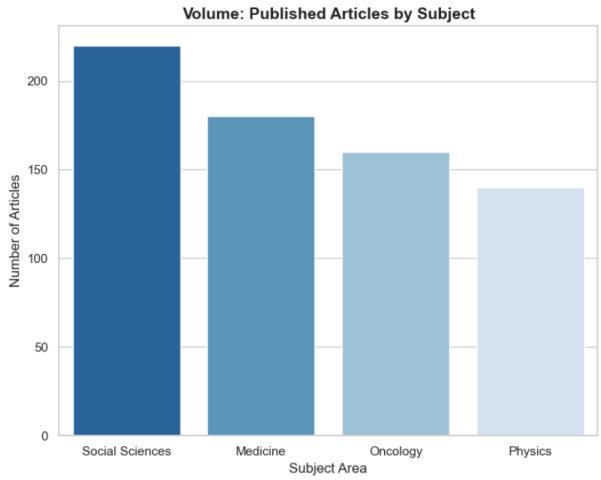
*Four cleaned DataFrames were exported as CSV files for MySQL ingestion, forming the Silver Layer - standardised, validated, and ready for the attribution model.*

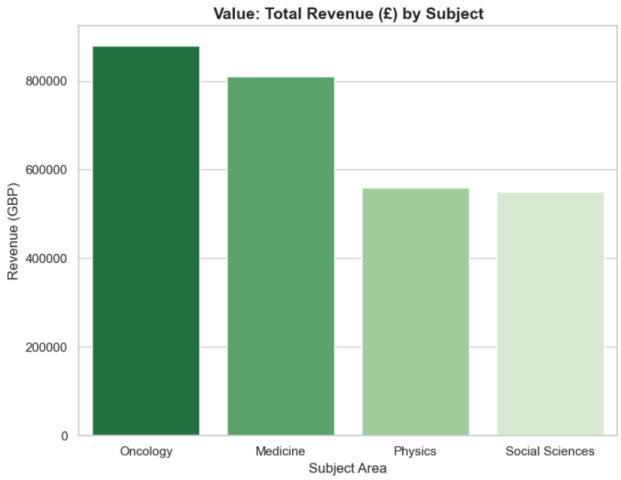
## 5.2 Exploratory Data Analysis

EDA was conducted across three analytical dimensions to surface initial growth opportunities and test hypotheses before building the SQL attribution model. Each dimension produced a chart and a business takeaway that informed subsequent analytical decisions.

### Chart 1 - Subject Area Volume vs Revenue

Published articles and APC revenue were compared across the four subject areas. Social Sciences produced the highest submission volume but the lowest revenue, while Oncology produced the lowest volume but the highest revenue - driven by its premium £5,500 APC rate.

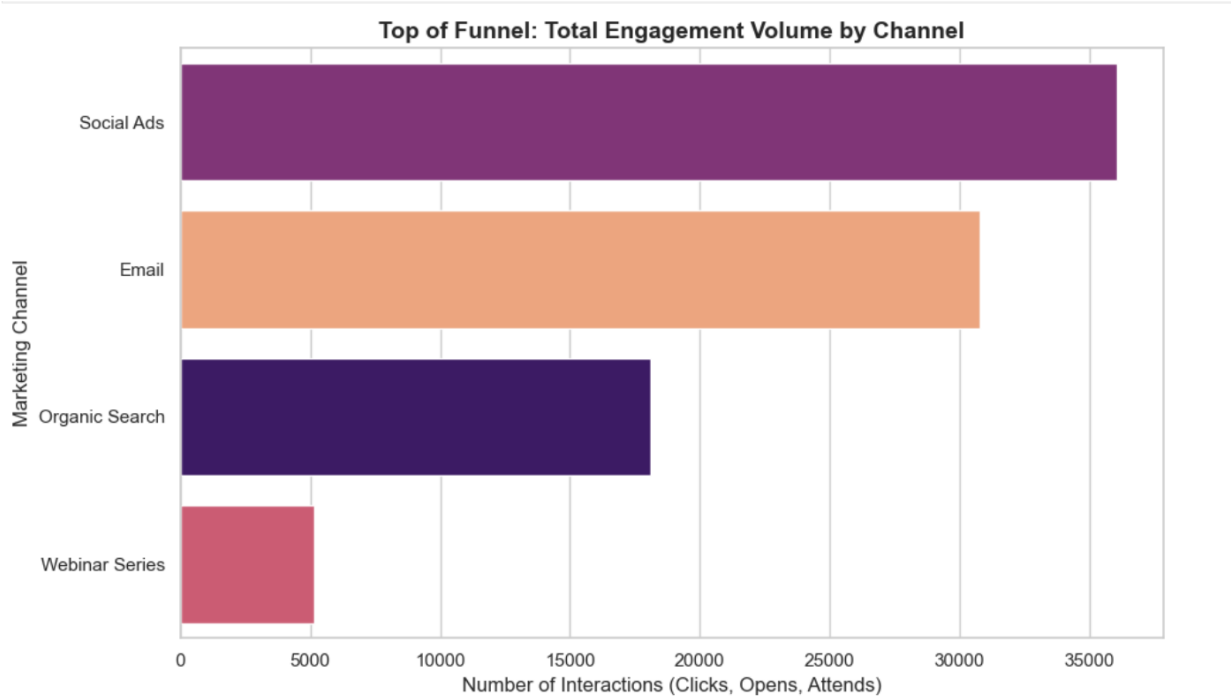



**Business Takeaway:** Marketing budget is being allocated toward high-volume but low-yield subject areas. Future campaigns should prioritise Oncology and Medicine researchers to maximise revenue per acquisition.

## Chart 2 - Top of Funnel: Engagement Volume by Channel

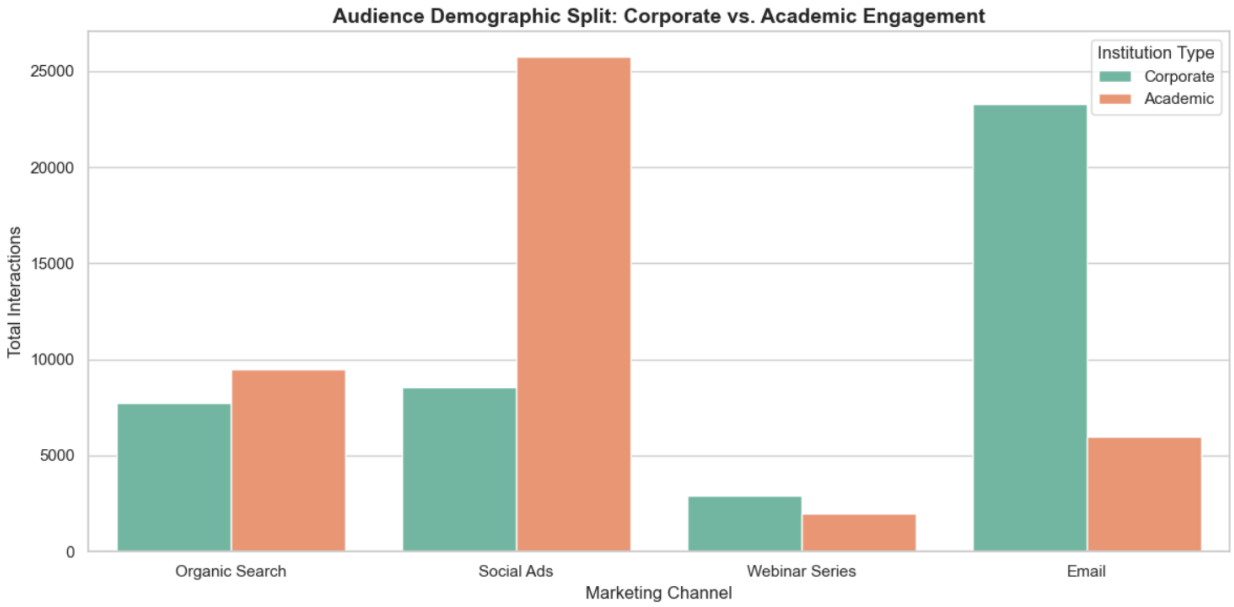
Raw interaction volume was profiled across all four channels. Social Ads generated the highest top-of-funnel volume with 36,052 interactions - more than any other channel. However, high volume was explicitly flagged as a potential vanity metric: it reveals nothing about conversion quality or revenue return.



**Business Takeaway:** Budget decisions cannot be made on traffic volume alone. A full attribution model (built in SQL) is required to calculate the true CPA and ROAS of each channel - the EDA finding that Social Ads leads on volume was the direct motivation for the SQL attribution phase.

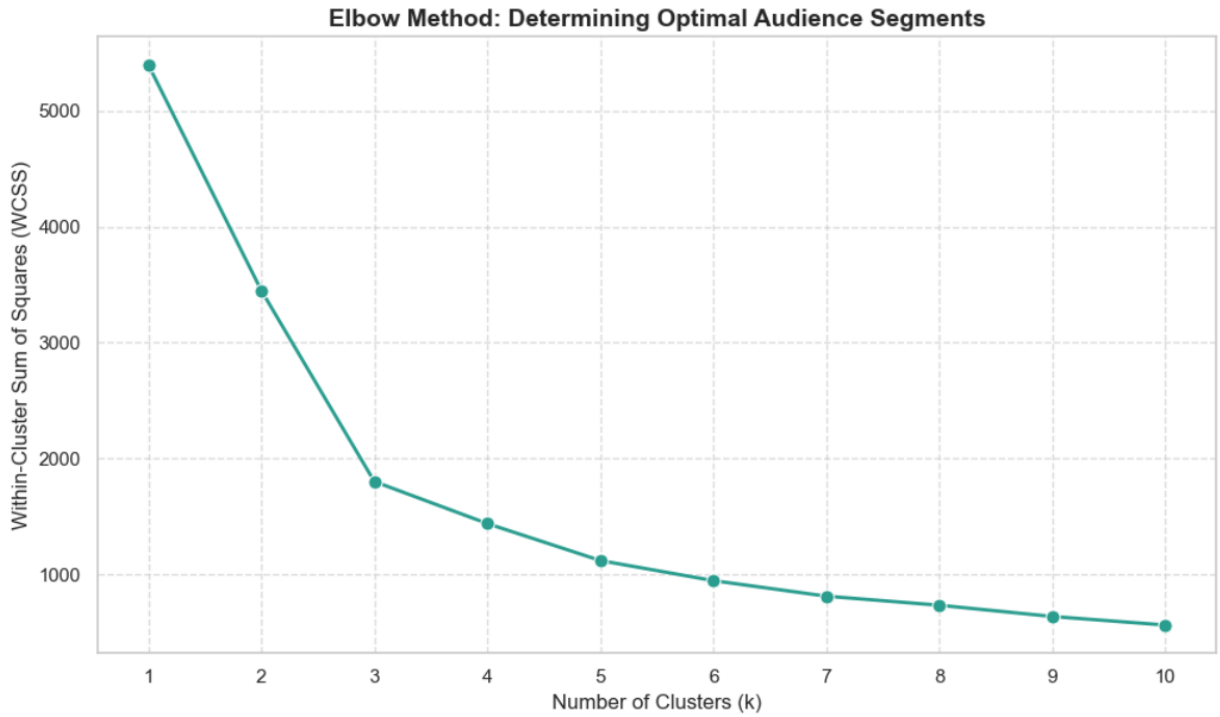
### Chart 3 - Channel Preference by Audience Persona

Marketing interactions were joined with author demographic data to compare channel behaviour between Corporate and Academic author personas. The analysis confirmed the hypothesis that these two groups are not a monolith - Corporate researchers exhibit distinctly different channel engagement patterns compared to Academic researchers, with Corporate authors showing higher email engagement intensity.

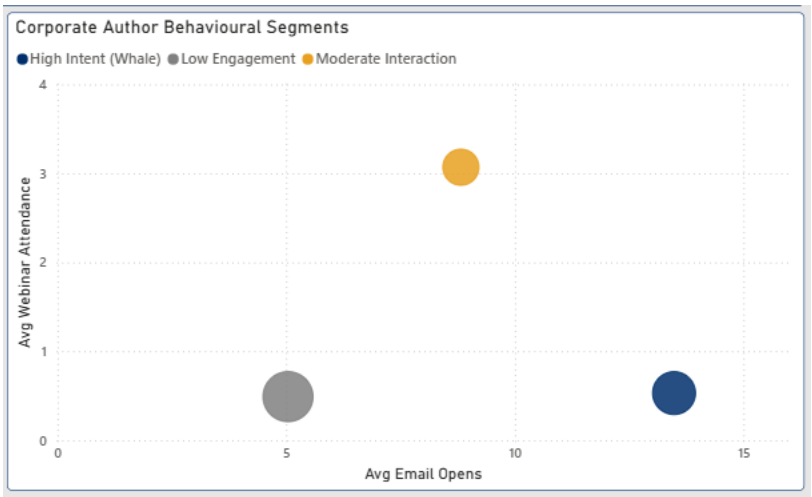



**Business Takeaway:** The behavioural divergence between Corporate and Academic authors confirmed that a one-size-fits-all campaign strategy is sub-optimal. This finding directly motivated the decision to apply K-Means clustering exclusively to corporate authors in the next phase, rather than across the full author population.

## 5.3 K-Means Clustering Results




| Cluster | Label | Avg Email Opens | Avg Webinar Attendance | Users | Dominant Segment |
|---|---|---|---|---|---|
| 1 | High Intent (Whale) | 13.48 | 0.53 | 875 | ~78% Pharma Research Scientists |
| 2 | Moderate Interaction | 8.82 | 3.07 | 602 | Mixed - cross-channel engagers |
| 3 | Low Engagement | 5.04 | 0.49 | 1,222 | Lowest engagement on both channels |



**Cluster Insight:** The High Intent (Whale) cluster opens emails at nearly 2.7x the rate of the Low Engagement cluster (13.48 vs 5.04 avg opens). Demographic validation confirms approximately 78% are Corporate Pharma Research Scientists - the single most commercially valuable author segment. This directly informs the recommendation to target this segment via Email campaigns.

# 6. Phase 2 - SQL: Schema Design and Attribution Modelling

## 6.1 Database Schema

A normalised relational schema was designed in MySQL following third normal form (3NF). Each non-key attribute depends only on the primary key, eliminating data redundancy and ensuring referential integrity across all four core tables.

| Table | Primary Key | Foreign Keys | Purpose |
|---|---|---|---|
| authors | author_id | — | Author demographic and consent data |
| submissions | submission_id | author_id → authors | Article submission outcomes and APC revenue |
| interactions | interaction_id | author_id → authors, campaign_id → marketing_spend | Marketing touchpoint records |
| marketing_spend | campaign_id | — | Campaign budget allocations by channel |

## 6.2 Last-Touch Attribution Logic

The attribution model was implemented as a three-step SQL process. All pre-submission interactions per author were ranked by recency using ROW_NUMBER() partitioned by author_id. Rank 1 (most recent) was selected and joined to submissions to assign channel and campaign to each published article's revenue. The complete logic was then productionised as the submissions_attributed MySQL view.

Limit to 1000 rows

```sql
WITH LastTouchMapping AS (
    SELECT
        s.submission_id,
        s.apc_gbp as revenue,
        i.channel,
        i.campaign_id,

        -- Sorting: Ranking the clicks chronologically
        ROW_NUMBER() OVER(
            PARTITION BY s.submission_id
            ORDER BY i.interaction_date DESC,
            CASE i.interaction_type
                WHEN 'Conversion Touch' THEN 1
                WHEN 'Registration'     THEN 2
                WHEN 'Click'            THEN 3
                WHEN 'Open'             THEN 4
             END ASC
        ) as touch_rank

    FROM submissions s
    JOIN interactions i ON s.author_id = i.author_id

    -- Filter 1: We only care about papers that actually generated revenue
    WHERE s.status = 'Published'

      -- Filter 2: The marketing click MUST have happened BEFORE the paper was submitted
      AND i.interaction_date < s.submission_date
)
-- To show the results
SELECT * FROM LastTouchMapping WHERE touch_rank = 1;
```

## 6.3 Channel ROAS - Attribution Results

| Channel | Budget Spend | Attributed Revenue | ROAS | CPA | Conversions |
|---|---|---|---|---|---|
| Email | £50,000 | £1,428,500 | 28.57x | £142 | 352 |
| Webinar Series | £80,000 | £780,500 | 9.76x | £394 | 203 |
| Organic Search | £120,000 | £378,000 | 3.15x | £1,290 | 93 |
| Social Ads | £400,000 | £213,000 | 0.53x | £7,692 | 52 |

*Social Ads consumed 62% of the total budget (£400,000 of £650,000) but generated the lowest attributed revenue. Email consumed only 7.7% of the budget yet generated 51% of all attributed revenue.*

# 7. Phase 3 - Power BI: Data Model and Interactive Report

## 7.1 Data Model - Star Schema

The Power BI data model connects directly to the MySQL publishing_roi_audit database. A star schema was built with five relationships - four active and one inactive, enabling time intelligence via USERELATIONSHIP() in DAX.

| Relationship | Join Key | Status |
|---|---|---|
| interactions → authors | author_id | Active |
| interactions → marketing_spend | campaign_id | Active |
| submissions → authors | author_id | Active |
| interactions → Date Table | interaction_date → Date | Active |
| submissions → Date Table | submission_date → Date | Inactive — activated via USERELATIONSHIP() |

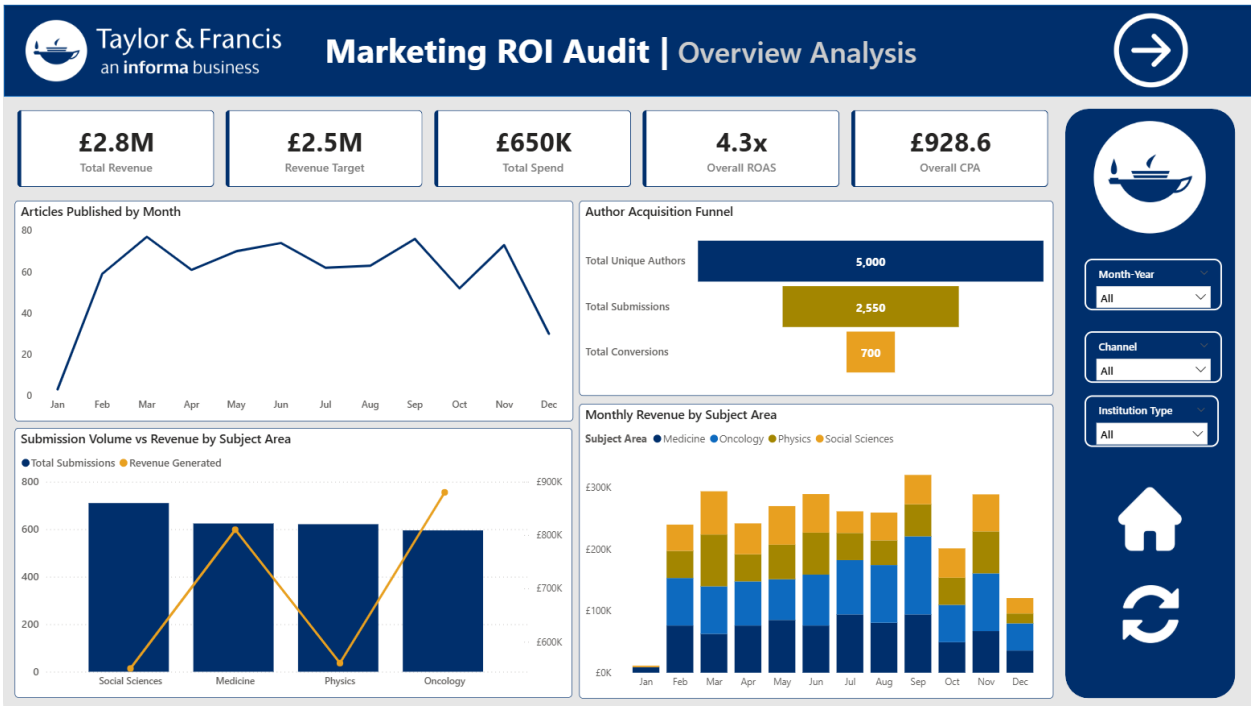
## 7.2 DAX Measures - 23 Measures in 6 Categories

| Folder | Measures |
|---|---|
| 01 - Core (6) | Total Revenue, Total Spend, Revenue Target, Total Conversions, Overall ROAS, Overall CPA |
| 02 - Funnel (3) | Total Interactions, Total Unique Authors, Total Submissions |
| 03 - Time Intelligence (2) | Monthly Revenue, Monthly Conversions - both use USERELATIONSHIP() |
| 04 - Channel Attribution (5) | Attributed Revenue, Channel Spend, Channel ROAS, Channel CPA, Channel Conversion Rate |
| 05 - Channel KPIs (3) | Best Channel ROAS, Worst Channel ROAS, Worst Channel CPA |
| 06 - Campaign Detail (4) | Campaign Budget, Campaign ROAS, Campaign CPA, Drill Title |

## 7.3 Report Pages

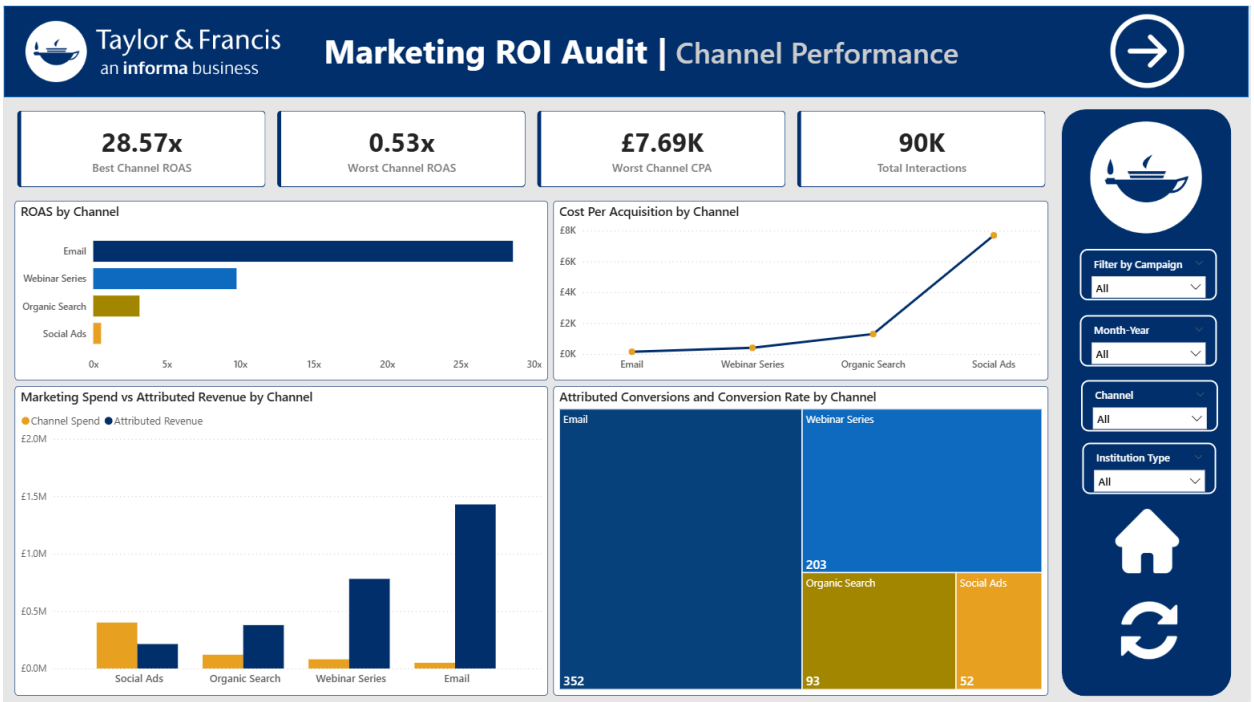
### Page 1 - Overview Analysis

Three global slicers (Month-Year, Channel, Institution Type) synced across Pages 1–3. Five KPI cards showing Total Revenue, Revenue Target, Total Spend, Overall ROAS, and Overall CPA. Four charts: monthly published articles line chart, author acquisition funnel, submission volume vs revenue by subject area combo chart, and monthly revenue by subject area stacked column chart.



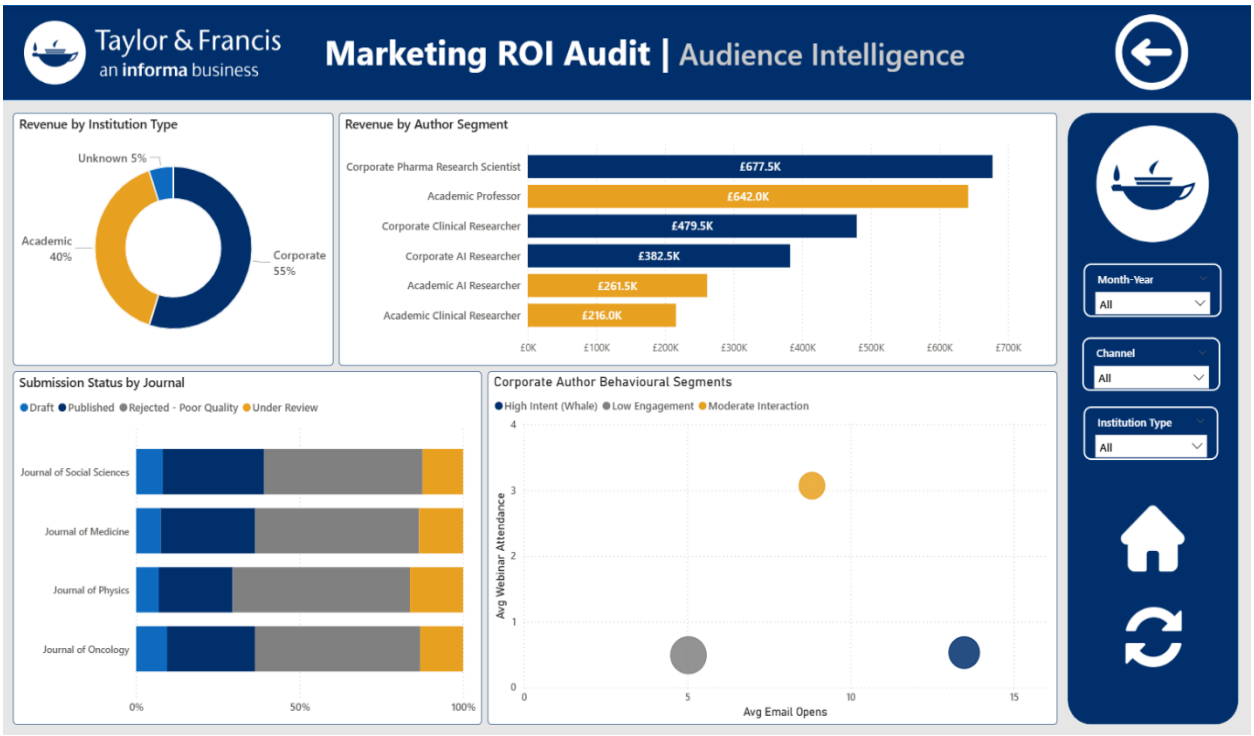

## Page 2 - Channel Performance

Four KPI cards: Best Channel ROAS (28.57x), Worst Channel ROAS (0.53x), Highest CPA (£7,692), Total Interactions (90,090). Four charts: ROAS by channel horizontal bar chart, spend vs revenue clustered column chart, CPA by channel line chart, attributed conversions treemap. Channel slicer Edit Interactions set to None on the treemap to prevent single-box collapse.



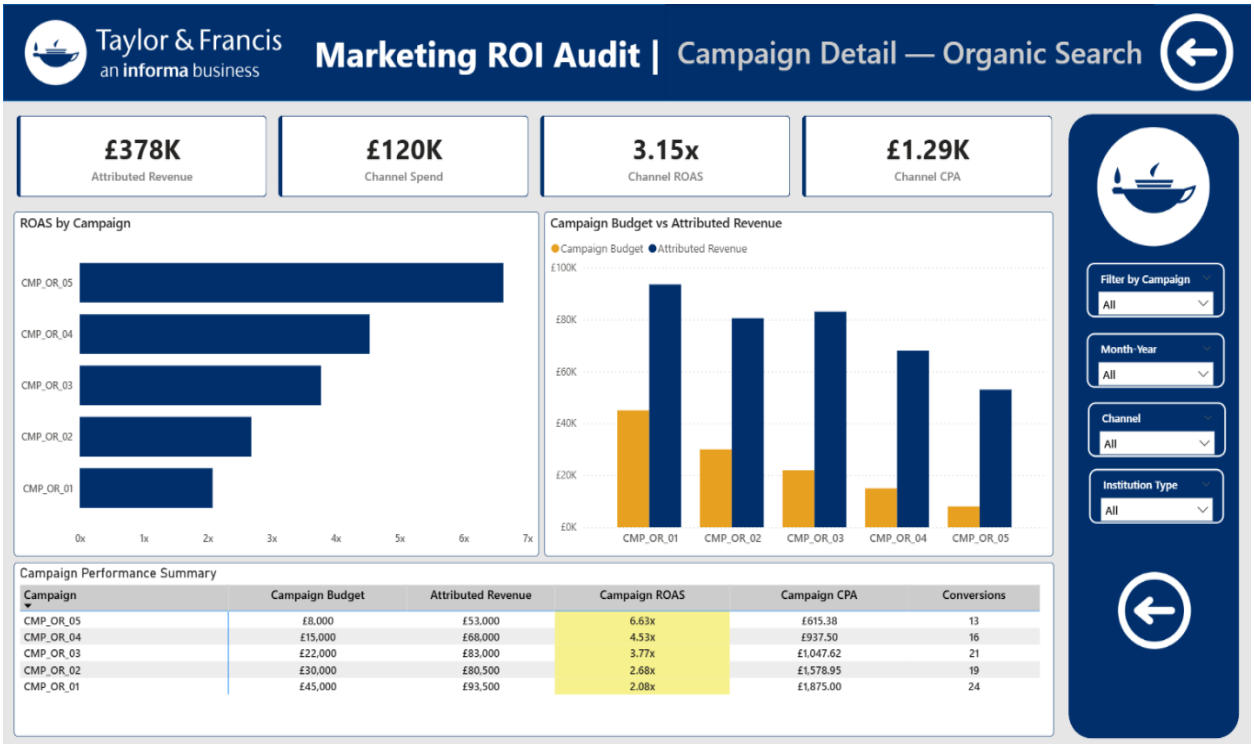
## Page 3 - Audience Intelligence

Four charts: revenue by institution type donut (Corporate 55%, Academic 40%, Unknown 5%), revenue by author segment clustered bar (Unknown filtered out, 6 segments), K-Means behavioural scatter plot, and submission status by journal 100% stacked bar chart.




## Page 4 - Campaign Detail (Hidden Drill-Through)

Accessed by right-clicking any channel bar on Page 2 and selecting Drill through. Drill-through field: Submissions Attributed[last_touch_channel]. Dynamic title card updates to show selected channel. Four KPI cards showing channel-level metrics. ROAS by campaign bar chart. Campaign budget vs revenue clustered column chart. Campaign performance matrix with conditional formatting: green (ROAS ≥ 10), amber (ROAS 2–10), red (ROAS < 1).




Campaign Performance Summary

| Campaign | Campaign Budget | Attributed Revenue | Campaign ROAS | Campaign CPA | Conversions |
|---|---|---|---|---|---|
| CMP_OR_05 | £8,000 | £53,000 | 6.63x | £615.38 | 13 |
| CMP_OR_04 | £15,000 | £68,000 | 4.53x | £937.50 | 16 |
| CMP_OR_03 | £22,000 | £83,000 | 3.77x | £1,047.62 | 21 |
| CMP_OR_02 | £30,000 | £80,500 | 2.68x | £1,578.95 | 19 |
| CMP_OR_01 | £45,000 | £93,500 | 2.08x | £1,875.00 | 24 |

# 8. Key Findings and Business Insights

## 8.1 Channel Performance

### Email - Outstanding ROI

Email delivered a 28.57x ROAS on £50,000 spend, generating £1,428,500 in attributed revenue from 352 published articles. With a CPA of just £142 per article, Email is the most efficient channel in the portfolio by every metric. This dominance is reinforced by the K-Means findings - the High Intent Whale cluster (875 users, ~78% Pharma Research Scientists) are predominantly Email-responsive with 13.48 average opens.

### Social Ads - Critical Budget Misallocation

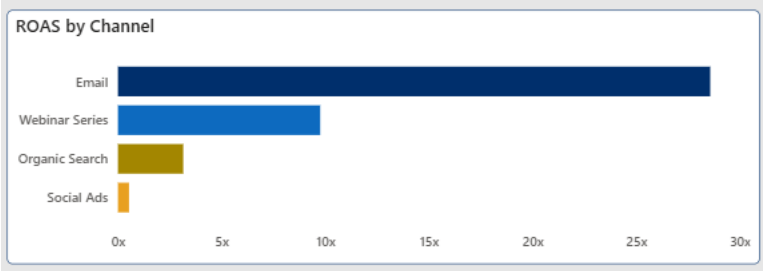
Social Ads consumed £400,000 - 62% of the total marketing budget, but delivered only £213,000 in attributed revenue, a 0.53x ROAS representing a net loss of £187,000. At £7,692 per published article, Social Ads CPA is 54x higher than Email. At campaign level, only one of five Social Ads campaigns (CMP_SO_05) broke even at 1.26x ROAS, the remaining four are all loss-making.

### Webinar Series - Strong and Underinvested

Webinar Series achieved a 9.76x ROAS on £80,000 spend, generating £780,500 from 203 published articles. It has the highest conversion rate of all channels at 3.95% and is the preferred channel for the Moderate Interaction cluster (602 users, 3.07 avg webinar attendance). It is significantly underinvested relative to its performance.

### Organic Search - Positive but Inefficient

Organic Search returned 3.15x ROAS on £120,000 spend - above break-even, but with a CPA of £1,290 it is significantly less efficient than Email and Webinar Series. Campaign-level analysis shows diminishing returns, the largest campaign (CMP_OR_01 at £45,000) produced the lowest per-£ return of the five Organic Search campaigns.



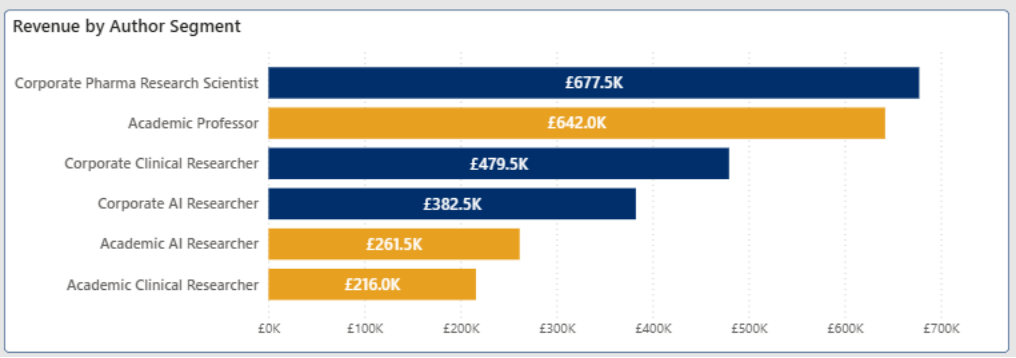

## 8.2 Audience Intelligence

### Institution Type Revenue Split

Corporate authors generate 55% of total revenue (£1,539,500). Academic authors contribute 40% (£1,119,500). The 5% Unknown segment (£141,000) represents a data quality gap - 250 authors with unclassified institution types across 39 published articles.

### Most Valuable Author Segment

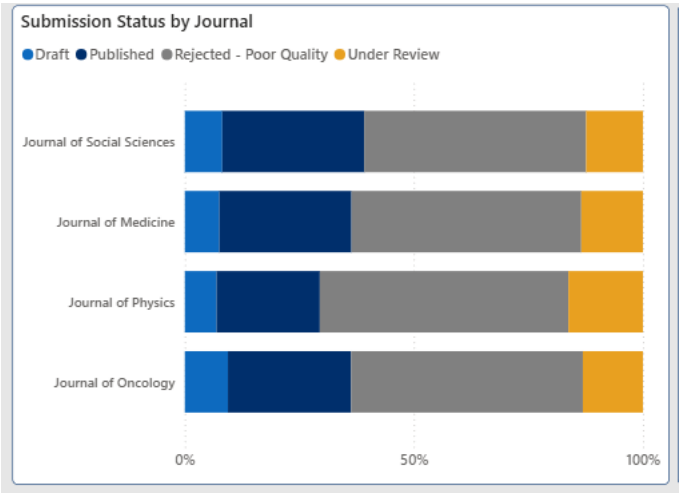
Corporate Pharma Research Scientists are the single most valuable author segment at £677,500 in attributed revenue, followed by Academic Professors at £642,000. Corporate authors dominate the top four revenue positions across all job title and institution combinations. This finding directly connects to the Whale cluster, the highest-value segment is also the highest email-engaging segment.




## 8.3 Subject Area and Journal Analysis

Oncology generates the highest revenue (£880,000) despite having the lowest submission volume (595), driven by its £5,500 APC rate. Social Sciences has the highest submission volume (710) but the lowest revenue (£550,000) at £2,500 APC per article, an inverse relationship between volume and value that has direct implications for editorial pricing strategy.

Journal of Social Sciences has the highest acceptance rate at 31% (220 of 710 submissions published). Journal of Physics has the lowest at 23% (140 of 621), with a rejection rate of 55% - the most selective journal in the portfolio. Under Review proportions are consistent across all four journals (13–16%), suggesting the differentiation in outcome is driven entirely by editorial standards.



# 9. Results and Recommendations

The audit confirms that the current marketing budget allocation is significantly misaligned with channel performance. The following five recommendations are presented in priority order based on potential revenue impact.

### Recommendation 1 - Reallocate Social Ads Budget Immediately

Social Ads is the only loss-making channel at 0.53x ROAS, representing a £187,000 net loss on £400,000 spend. Reducing this allocation by at least 50% (£200,000) and redistributing to Email (£100,000) and Webinar Series (£100,000) would be expected to materially increase overall ROAS. If Email's historical 28.57x rate were applied to an additional £100,000 of spend, projected incremental revenue would approach £2,857,000 - though actual returns at higher scale may be lower due to audience saturation.

### Recommendation 2 - Prioritise Corporate Pharma Research Scientists

Future Call for Papers (CFP) campaigns should be targeted specifically at Corporate Pharma Research Scientists - the dominant demographic of the High Intent Whale cluster. Email is the confirmed channel for this segment. Campaigns should be personalised to Oncology and Medicine subject areas where Pharma authors generate the highest APC revenue.

### Recommendation 3 - Activate Moderate Interaction Cluster via Webinar

The 602-user Moderate Interaction cluster shows high webinar attendance (3.07 avg) but only moderate email engagement. Dedicated webinar campaigns targeting this segment are expected to yield higher conversion rates than generic email outreach, leveraging the channel that this cluster demonstrably prefers.

### Recommendation 4 - Journal Submission Guidance

Authors should be guided toward journals that match their submission quality to maximise acceptance rates. Journal of Social Sciences (31% acceptance) is appropriate for broader audience outreach campaigns. Journal of Physics (23% acceptance, 55% rejection) requires more targeted outreach to high-quality submissions to justify its selectivity.

### Recommendation 5 - Resolve Unknown Institution Type Records

250 authors with Unknown institution type represent a data quality gap that limits segmentation accuracy. Resolving these records through CRM enrichment or email verification would improve attribution quality and enable more precise audience targeting in future reporting periods.

## Results Summary

| Metric | Current State | Recommended Action |
|---|---|---|
| Social Ads ROAS | 0.53x (net loss £187K) | Reduce spend by 50%, reallocate to Email |
| Email ROAS | 28.57x on £50K | Increase budget allocation significantly |
| Webinar ROAS | 9.76x on £80K | Increase budget, target Moderate Interaction cluster |
| Primary Target Segment | All authors broadly | Corporate Pharma Research Scientists via Email |

| | | |
|---|---|---|
| Journal Strategy | No guidance | Direct authors to Social Sciences (31% acceptance) |